



PhotoVault

Your photos on Amazon Photos or OneDrive. The cloud only sees noise.

Free · Open source · No ads · No tracking · Donation-funded

github.com/rimaturus/PhotoVault · v1.5 beta · Android 13+

THE PROBLEM

Cloud photos are read, not just stored

Scanned

Faces, places, objects and text in your pictures are analysed and indexed as soon as they are uploaded.

Reused

That analysis can feed search, advertising profiles and the training of AI models under terms you accepted once.

Exposed

A breach, a leaked password or a hijacked account gives someone else your whole photo history, with GPS and dates.

And yet the space is often already paid for: unlimited photos with Amazon Prime, OneDrive with Microsoft 365. The space is great; handing over the content is the problem.

THE IDEA

Keep the storage. Remove the content.

What you see in PhotoVault



sample.jpg, 72 KB, decrypted on your phone

What Amazon stores



random-name.png, 156x155 px of ciphertext

Each photo is encrypted on the phone and uploaded as a valid PNG made of random pixels. Only PhotoVault, with your password, turns it back into your photo.

HOW IT WORKS

Encrypted on the phone, before it leaves

1 Pick

Android photo picker:
the app sees only
what you choose



2 Encrypt

AES-256-GCM with a
key derived from your
password



3 Pack

Ciphertext becomes
the pixels of a PNG
with a random name



4 Upload

Only that PNG goes to
your Amazon Photos
or OneDrive

Viewing is the same path backwards

Download the PNG, verify that not a single bit changed (the GCM tag), decrypt in memory. A small encrypted preview on the phone shows up instantly while the original downloads.

Standard, boring, well-tested cryptography

Key derivation	PBKDF2-HMAC-SHA256, 600,000 iterations (OWASP), random 16-byte salt
Encryption	AES-256-GCM, fresh random nonce for every file
Integrity	128-bit GCM tag over data and header: tampering or a wrong key is rejected
Metadata	File name, date and EXIF are inside the encrypted part
Implementation	Android's built-in crypto only: no custom cipher, no third-party library
Two implementations	Java (app) and Python (PC recovery tool), cross-tested byte for byte

The key exists only on your phone. Nobody can reset your password: not Amazon, not Microsoft, not the developer.

What the cloud can and cannot see

Amazon / Microsoft can see

- That you upload PNG files of random noise
- How many, how big, and when
- That they are probably encrypted
- Your account, as with any upload

Amazon / Microsoft cannot see

- ✓ Any content of your photos and videos
- ✓ Faces, places, objects, text
- ✓ File names, dates, GPS, camera, any EXIF
- ✓ Your vault password or key

There is nothing to scan and nothing to train AI on: only uniform random bytes.

Hardened by default

- **Key in memory only**
Wiped on lock. Auto-lock 60 s after you leave the app.
- **Fingerprint unlock**
Key wrapped by a hardware-backed Android Keystore key.
- **Everything local is encrypted**
Item list, previews and cached downloads.
- **Decrypted video**
Exists on disk only while you watch it.
- **No backups, no transfers**
Android backup and device-to-device copy disabled.
- **No screenshots**
Blank in recent apps (can be allowed temporarily).
- **Safe sign-in**
Amazon: only its sign-in sites. OneDrive: Microsoft's page, OAuth + PKCE, app folder only.
- **5 permissions only**
No storage, contacts, location, camera or microphone.

Honest about what it protects

Protects your photos against

- ✓ The provider scanning or AI training
- ✓ A breach of the cloud storage
- ✓ Someone getting into your cloud account
- ✓ Silent modification of stored files

Does not protect against

- ✗ A weak password (offline guessing)
- ✗ Malware on an unlocked phone
- ✗ Metadata: count, size, timing
- ✗ Losing your password: no recovery, by design
- ✗ The provider deleting files: keep a backup

No independent audit yet. Code review and reports are welcome.

OUR PROMISE

Your photos are not the product

- **No ads** Not now, not later.
- **No tracking** No analytics, no crash reporting, no SDKs. Zero third-party libraries.
- **No data selling** No servers, no accounts of ours. We never receive anything, so there is nothing to sell or leak.
- **Open source** GPL-3.0. Anyone can read, build and verify the whole app.
- **Donation-funded** Free for everyone. Donations are voluntary and unlock nothing.

FEATURES

Built for everyday use

Background uploads

Add many photos at once and leave: a notification shows progress and has a Stop button.

Instant previews

Tap a photo: its preview appears at once, the original follows. Swipe to the next.

Folders

Organise photos in folders. Their names are encrypted too, and restored on a new phone.

New phone? Sync

Sign in, same password: the vault is rebuilt from the cloud.

Self-test

8 checks on your own account before you store anything.

Amazon or OneDrive

Unlimited photos with Prime, or OneDrive through Microsoft's official API.

BY THE NUMBERS

Small enough to read, fast enough to forget

0

third-party libraries

0

servers, accounts, trackers

280

lines of crypto code

+0.3 %

size overhead (0.5 MB photo)

600k

PBKDF2 iterations

~30 ms

to encrypt a photo*

* Measured on one desktop CPU core. The in-app self-test shows the real timings on your phone.

Limits and risks, in plain words

! **Amazon: unofficial API**

No public Photos API: PhotoVault uses the website's own requests. If Amazon changes them, uploads pause until an update.

! **Amazon: videos**

Prime is unlimited for photos, 5 GB for videos. Encrypted videos stored as images go against the spirit of the offer.

! **OneDrive: your quota**

Files count against your OneDrive storage (5 GB free, 1 TB with Microsoft 365).

! **Password = only key**

It can't be recovered. Changing it re-encrypts the whole vault in the background.

! **Beta**

Not independently audited yet. No Google Photos: its API can't store or return exact files.

GET INVOLVED

Free, open, and supported by people like you

Get it

- APK and source:
github.com/rimaturus/PhotoVault
- Android 13 or newer
- Recover anytime with photovault.py
- Report bugs, review the code, share it

♥ Support PhotoVault

No ads, no tracking, no paid tier: ever. If PhotoVault is useful to you, a voluntary donation funds development and, one day, an independent security audit.

buymeacoffee.com/rimaturus

Also in the app: Settings > Support PhotoVault

Donations unlock nothing: everything stays free for all.

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